

Barriers to artificial intelligence adoption in State-Owned Enterprises.

Patience Mokokota*,

***Corresponding Author:** Student, Applied Information Systems, University of Johannesburg, 37 Bunting Rd, Cottesloe, Johannesburg, 2092. E-mail: patiencemokokota@gmail.com
ORCID:
<https://orcid.org/0009-0007-9351-2417>

Shopee Dube,

Senior Lecturer, Applied Information Systems, University of Johannesburg, 37 Bunting Rd, Cottesloe, Johannesburg, 2092. E-mail: shopeed@uj.ac.za
ORCID:
<https://orcid.org/0000-0003-0033-7992>

Tinofrei Museba,

Lecturer, Applied Information Systems, University of Johannesburg, 37 Bunting Rd, Cottesloe, Johannesburg, 2092. E-mail: tmuseba@uj.ac.za
ORCID: <https://orcid.org/0000-0003-0356-5245>

Abstract

State-owned enterprises (SOEs) have significant opportunities to enhance operational efficiency, decision-making, and service delivery through artificial intelligence (AI). However, several barriers hinder widespread AI adoption in these organisations. This article measures the impact of AI adoption on performance indicators like production output and customer satisfaction in state-owned enterprises. It analyses financial sustainability, resource management, and efficiency metrics while identifying key barriers such as high implementation costs, lack of technical expertise, and resistance to change. The barriers to AI adoption in the SOEs are studied quantitatively. Data was gathered from 240 carefully chosen SOE personnel in South Africa (SA) using a survey approach, which had a 93.28% response rate. The population of the article is a sample of SOE employees. The focus of the article is on examining the perspectives of individuals in SOEs in SA regarding AI implementation. Further complicating AI integration are data security and privacy concerns, as well as legal and compliance issues. A lack of strategic vision and financial limitations are also major obstacles to advancement. Developing successful plans to encourage AI adoption in SOEs, which will eventually spur innovation and enhance public sector performance, requires an understanding of these obstacles. The article concludes that despite significant barriers to AI adoption in South African SOEs—such as lack of senior management support, inadequate infrastructure, resistance to change, data quality issues, and a shortage of skilled staff—there are clear opportunities for improvement. This is one of the articles that focuses on identifying barriers to adopting AI in South African SOEs, and it lays a foundation for future research on AI in SOEs.

Keywords: Artificial Intelligence, State-Owned Enterprises, Barriers, AI Adoption.

JEL Classification: L32, O33, M15.

Introduction

The adoption of AI by SOEs in SA has faced significant hurdles in recent years, inhibiting its widespread implementation and potential benefits. One of the primary obstacles is the financial constraint many SOEs experience, with limited budgets preventing investments in innovative technologies. High implementation costs further compound these financial challenges (Shang et al., 2023). Additionally, the shortage of AI-trained professionals presents a critical bottleneck, as these organizations struggle to recruit and retain skilled workers capable of managing and operating AI systems effectively.

Cultural resistance within SOEs also plays a crucial role, as employees often oppose AI-driven advancements out of fear of job displacement (Kar & Kushwaha, 2021). Moreover, strict regulatory and compliance requirements present another layer of complexity, complicating the integration of modern technologies into these enterprises (Radhakrishnan & Chattopadhyay, 2020). Concerns about data security and privacy are particularly pronounced in the public sector, where the handling of sensitive information demands robust safeguards.

Despite AI's transformative potential, various structural, cultural, and technical challenges hinder its adoption. The perception that information technology (IT) is insufficiently mature, along with fears of relinquishing control to machine-based algorithms, remains a pervasive barrier (Edmondson, 2023). Organisational readiness, including the availability of skilled personnel, adequate infrastructure, and a willingness to embrace change, is another critical factor influencing adoption rates (Rajput, 2023). The prohibitive costs of implementation and maintenance, combined with limited top-down support and a scarcity of AI-trained staff, exacerbate these challenges (Shang et al., 2023).

Interestingly, while the demand for AI and analytics expertise continues to grow, the skills gap persists, leaving many organizations struggling to meet this demand (McKendrick, 2021). Unlocking the potential of AI adoption in SOEs requires comprehensive top-down support, sufficient resource allocation, and a focus on building organizational preparedness in terms of both financial and human capital. Addressing these issues is imperative for South African SOEs to leverage AI's benefits effectively while overcoming political pressures, legislative restrictions, and administrative roadblocks. The article is essential because it clarifies the difficulties that these businesses have, including political pressures, legislative limitations, and administrative roadblocks. As a result, the study intends to accomplish the following objective:

- To measure the impact of AI adoption on performance indicators like production output and customer satisfaction in SOE, analyze the financial sustainability, resource management, and efficiency metrics while identifying key barriers such as high implementation costs, lack of technical expertise, and resistance to change.

Literature review

The adoption of AI in SOEs is being widely discussed in the literature, with several key obstacles emerging as critical barriers to implementation. Financial challenges, such as the high costs associated with installing and maintaining AI systems, are often cited as major constraints. Another prominent issue is the significant skills gap, with a shortage of employees possessing the requisite expertise in AI technologies. Resistance to organizational change further exacerbates these challenges, frequently rooted in limited awareness of AI's potential benefits and fears of job displacement among employees. Addressing these multifaceted obstacles and successfully embedding AI in SOEs demands a clear strategic vision and a strong commitment from leadership. This review explores the various barriers identified in the literature and emphasizes the importance of organizational preparedness in overcoming these hurdles.

Telkom, Transnet, and Eskom are important South African SOEs that have received recognition for their AI initiatives (Marrie et al. 2023). Transnet employs AI to optimize coordination and supply chain management, improving efficiency and decision-making. Telkom has integrated AI into its business platform, enhancing services for small and medium-sized enterprises (SMEs) (Gwabeni 2023). Eskom uses AI for predictive maintenance and customer service through an AI Chatbot named Alfred, which helps manage customer interactions and report outages (Ratshalingwa 2023).

Despite these advancements, these SOEs face significant challenges, including financial constraints, cultural resistance to change, infrastructure modernization needs, skills shortages, and regulatory compliance issues. Addressing these challenges is crucial for maximizing the benefits of AI and driving innovation within South African SOEs. The state-owned logistics and freight transportation firm Transnet in SA has been investigating the use of AI to improve its business processes (Souček & Lokoč 2020). The goal of integrating AI technology is to enhance decision-making, safety, and efficiency in a variety of areas of their services. To improve its operations, Transnet has been aggressively using AI technology (Souček et al. 2019). Figure 1 presents a conceptual model illustrating the key barriers and factors influencing the adoption of AI technologies within SOEs.

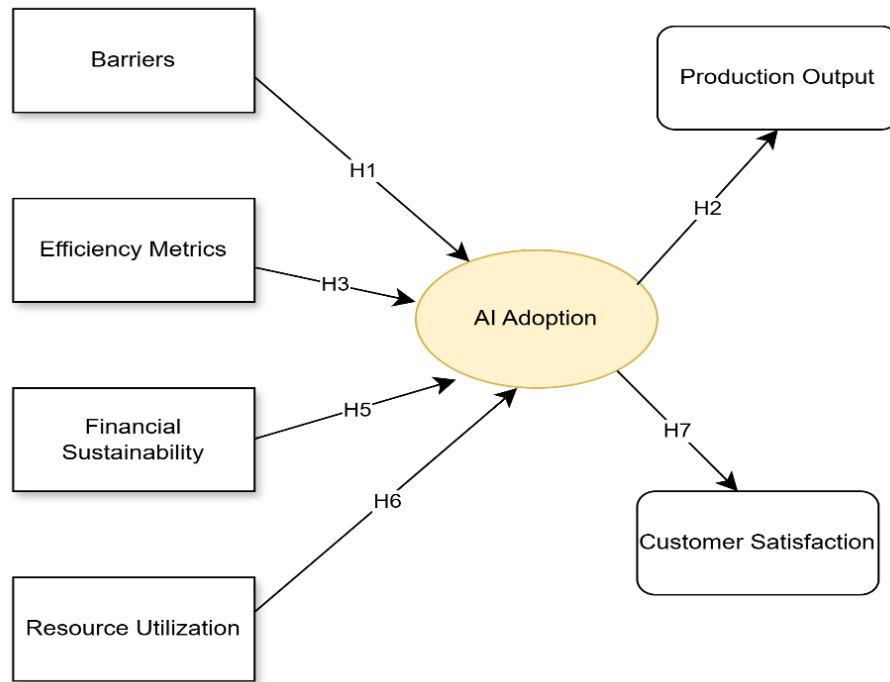


Figure 1: Conceptual Model of the Study; *Source:* Authors

Barriers

Barriers to the adoption of AI within SOEs constitute a significant challenge, encompassing a multitude of impediments that obstruct the effective deployment of AI technologies. These impediments encompass regulatory complexities, characterized by intricate and stringent policies, which frequently decelerate or obstruct advancement, thereby corroborating the sub-hypothesis (**H1a**) that heightened regulatory complexity exhibits a negative correlation with AI adoption. An additional considerable barrier is the deficiency of technical expertise within SOEs, as limited access to proficient personnel frequently constrains their capacity to implement and manage AI technologies effectively, thereby aligning with sub-hypothesis (**H1b**) that a dearth of expertise results in diminished AI adoption rates. Resistance to organizational change, arising from entrenched cultural norms and employee perceptions, further undermines adoption initiatives, as certain stakeholders harbor fears of job displacement or lack comprehension of the advantages of AI, thereby supporting the sub-hypothesis (**H1c**) that pronounced resistance within organizations correlates with reduced adoption rates. Financial constraints also manifest as a pivotal barrier, as budgetary limitations curtail investment in the essential infrastructure, training, and tools requisite for AI implementation, reflecting the **H1d** that financial constraints adversely affect AI adoption.

Collectively, these interrelated barriers substantiate the overarching hypothesis (**H1**): the greater the magnitude of the barriers, the lower the rate of AI adoption in SOEs, thereby emphasizing the necessity for strategic interventions to alleviate these challenges and promote the successful integration of AI.

Production Output

Production output, which serves as a metric for evaluating the quantity and quality of goods or services generated by SOEs, is integral to the assessment of organizational efficacy. The integration of AI possesses substantial potential to refine production methodologies, yielding notable advancements in both the qualitative and quantitative dimensions. The sophisticated functionalities of AI in the realms of process regulation and accuracy can significantly enhance the caliber of products or services, thereby substantiating the sub-hypothesis (**H2a**) that the implementation of AI engenders considerable enhancements in product/service quality. Furthermore, the automation and operational efficiency propelled by AI facilitate organizations in scaling their production methodologies, thereby augmenting output levels and corroborating with sub-hypothesis (**H2b**) that the adoption of AI results in an escalation of production quantity.

Collectively, these nuanced elements of production output robustly substantiate the principal **H2** that the integration of AI exerts a positive influence on the production output of SOEs. These conclusions underscore the imperative for organizations to allocate resources towards AI technologies to bolster operational efficiency, elevate product quality, and enhance production capacity, thereby ensuring sustained competitiveness in the long term.

Efficiency Metrics

Efficiency metrics are critical indicators used to evaluate the performance of SOEs in areas such as time management, cost reduction, and resource utilization. The integration of AI technologies has the potential to streamline operations, minimize waste, and significantly enhance overall efficiency. AI adoption can improve time efficiency by automating repetitive tasks and optimizing workflows, supporting the sub-hypothesis (**H3a**) that AI adoption improves time efficiency in SOEs. Furthermore, AI-driven automation and operational optimization can lead to substantial cost savings, aligning with the sub-hypothesis (**H3b**) that AI adoption improves cost efficiency. Additionally, AI technologies enable better utilization of resources such as materials, energy, and workforce, further supporting the sub-hypothesis (**H3c**) that AI adoption enhances resource utilization efficiency.

Collectively, these detailed aspects support the main hypothesis (**H3**): higher efficiency metrics are associated with greater AI adoption in SOEs. This underscores the importance of leveraging AI technologies to achieve operational excellence and long-term sustainability within SOEs.

AI Adoption

AI adoption delineates the extent to which artificial intelligence technologies are executed and assimilated into the operational frameworks of SOEs, governed by crucial determinants such as organizational preparedness, resource availability, and the anticipated benefits. Organizational preparedness, which serves as an indicator of an SOE's readiness for the integration of AI, assumes a critical position in the adoption process, corroborating the sub-

hypothesis **(H4a)** that enhanced organizational preparedness correlates with elevated levels of AI adoption. The presence of requisite resources—financial, technological, and human—further facilitates the adoption trajectory, substantiating the sub-hypothesis **(H4b)** that an increased availability of resources exerts a positive influence on AI adoption. Moreover, the perceived utility and advantages associated with the adoption of AI technologies dictate decision-making processes, as suggested by the sub-hypothesis **(H4c)** that a heightened perception of benefits correlates with an escalation in AI adoption.

In aggregate, these dimensions bolster the principal hypothesis **(H4)**: organizational preparedness and resource availability exert a positive influence on AI adoption within SOEs. Addressing these determinants can engender a conducive environment for the successful integration of AI, thereby ensuring sustained efficiency and innovation within operational practices.

Financial Sustainability

Financial sustainability evaluates the long-term economic viability of SOEs, and AI adoption plays a pivotal role in enhancing this through cost savings, revenue generation, and overall financial health. AI technologies can significantly reduce operational and maintenance costs by automating processes and optimizing resources, supporting the sub-hypothesis **(H5a)** that AI adoption leads to significant cost savings. Additionally, AI-driven innovation enables organizations to create new revenue streams by introducing advanced products and services, aligning with the sub-hypothesis **(H5b)** that AI adoption enhances revenue generation. Moreover, AI can improve key financial metrics such as profitability, return on investment (ROI), and cash flow management, reflecting the sub-hypothesis **(H5c)** that AI adoption positively impacts overall financial health.

These aspects collectively support the main hypothesis **(H5)**: AI adoption enhances the financial sustainability of SOEs. By leveraging AI, SOEs can achieve long-term economic stability and competitiveness, ensuring their sustained growth and resilience in dynamic markets.

Resource Utilization

Resource utilization focuses on how effectively SOEs allocate and use their human, financial, and technological resources, and AI adoption has the potential to significantly enhance this efficiency. AI technologies can optimize workforce allocation, automate routine tasks, and improve overall productivity, supporting the sub-hypothesis **(H6a)** that AI adoption improves human resource utilization. Additionally, AI can streamline financial resource allocation, ensuring that budgets and investments are effectively directed toward high-value activities, aligning with the sub-hypothesis **(H6b)** that AI adoption improves financial resource utilization. Furthermore, AI enables SOEs to maximize the usage and efficiency of their technological assets by enhancing system integration, reducing downtime, and improving

functionality, reflecting the sub-hypothesis **(H6c)** that AI adoption enhances technological resource utilization.

Collectively, these aspects support the main hypothesis **(H6)**: improved resource utilization is linked to higher AI adoption in SOEs. By leveraging AI to optimize these resources, SOEs can achieve greater operational efficiency, cost-effectiveness, and sustainable performance outcomes.

Customer Satisfaction

Customer satisfaction reflects the extent to which customers are pleased with the products or services offered by SOEs, and AI adoption has significant potential to enhance this critical metric. AI technologies can improve the quality of customer service and support by providing faster, more accurate, and reliable assistance, supporting the sub-hypothesis **(H7a)** that AI adoption enhances service quality. Additionally, AI enables organizations to deliver personalized experiences through advanced data analytics and tailored solutions, aligning with the sub-hypothesis **(H7b)** that AI adoption improves the personalization of services. Furthermore, AI-driven tools, such as engagement platforms, foster better interaction and communication with customers, supporting the sub-hypothesis **(H7c)** that AI adoption increases customer engagement.

These elements collectively validate the main hypothesis **(H7)**: AI adoption leads to higher customer satisfaction in SOEs. By leveraging AI to improve service quality, personalization, and engagement, SOEs can not only meet but exceed customer expectations, ensuring greater loyalty and competitive advantage in their respective sectors.

Research methods and design.

To address research questions or test hypotheses, research methodology is a methodical and scientific approach to gathering, evaluating, and interpreting quantitative or qualitative data (Sreekumar, 2023). This article used a quantitative research approach, focusing on using structured questionnaires to gather data from SOE employees. The research design included descriptive analysis to identify patterns and correlations.

Sample and sample size.

Market research professionals mostly employ probability and non-probability sampling techniques. In this case, probability sampling was used, where a sample is randomly selected from a larger population, ensuring each member has an equal chance of being chosen. (Shin, 2024). Data was gathered from 240 carefully chosen SOE personnel in SA using a survey approach, which had a 93.28% response rate. The population of the article is a sample of SOE employees. The focus of the article is on examining the perspectives of individuals in SOEs in SA regarding AI implementation.

Data analysis

The article's data analysis and findings concerning the deployment of AI in SA's SOEs are presented in this chapter. The main goal of this section is to analyze the data gathered to comprehend the attitudes, obstacles, and degrees of AI integration within the sampled SOEs. Descriptive statistics, such as mean, median, and standard deviation calculations, were performed using SPSS to provide a thorough picture of the obstacles to AI adoption and the present status of AI deployment in these enterprises. This section presents the statistical analysis and interprets the findings, offering insights into how AI is perceived and utilized in a sample of South African SOEs. It also highlights key areas for development to enhance productivity in these organizations. Here, the focus of the article used organized questionnaires to conduct a survey and get insights from SOE managers and employees.

Results

The expected outcomes include quantitative evidence on AI adoption in SOEs and insights into sector-specific challenges. Table 1 displays the findings of a reliability analysis for several constructs associated with the article using Cronbach's alpha.

Table 1: Cronbach's alpha – Reliability

No	Items Tested	Cronbach's Alpha Results
1	Barriers	0,797
2	Efficiency metrics	0,834
3	Financial sustainability	0,880
4	Resource utilization	0,935
5	Ai adoption	0,929
6	Production output	3
7	Customer satisfaction	237

Source: Authors

Among other constructions, it contains barriers, efficiency metrics, and customer satisfaction. Internal consistency is shown by Cronbach's alpha value for each construct—values above 0.80 are regarded as acceptable, and those above 0.90 as exceptional. The Cronbach's Alpha results provided in Table 1 indicate the reliability of the scales used to measure each construct in the article. Cronbach's alpha values above 0.7 are considered acceptable, indicating good internal consistency. The level of AI adoption in the SOEs is displayed in Table 2.

Table 2: Level of AI implementation in the SOE organizations.

No		Level AI of Implementation	Frequency	Percent	Valid Percent	Cumulative Percent
1	Valid	Not yet started	73	30,4	30,8	30,8
2		Planning stage	73	30,4	30,8	61,6
3		Pilot phase	32	13,3	13,5	75,1
4		Partial implementation	48	20,0	20,3	95,4
5		Full implementation	11	4,6	4,6	100,0
	Missing		3	1.3		
	Total		237	98,8	100,0	

Source: Authors

Only 4.6% of the participating SOEs have fully implemented an AI project, 30.4% have not started any AI projects, 30.4% are still in the planning stage of AI implementation, 13.3% are still in the pilot stage, and 20% are presently in a partial implementation phase.

Descriptive statistics of the dataset

This is one of the most crucial tasks when doing statistical data analysis (Hassan, 2024b). The article investigates several variables affecting organizational effectiveness, as indicated by the descriptive data displayed in Table 3.

Table 3: Descriptive Statistics

No	Items Tested	Mean	Std. Deviation	N
1	Barriers	2,9634	,79399	196
2	Efficiency metrics	3,3867	,83170	196
3	Financial sustainability	3,5064	,84293	196
4	Resource utilization	3,5315	,85851	196
5	Ai adoption	2,9439	,84378	196

6	Production output	3,5262	,77604	196
7	Customer satisfaction	3,8280	,72669	196

Source: Authors

The descriptive statistics highlight key insights into the dataset and align with the proposed hypotheses. Barriers to AI adoption scored a moderate mean (2.9634), indicating their notable presence but varied perceptions among respondents, supporting **H1** that barriers negatively influence AI adoption, albeit with limited statistical significance. AI adoption itself also exhibited a moderate mean (2.9439), suggesting limited implementation, while production output (3.5262) and customer satisfaction (3.8280) showed higher means, reflecting the positive impacts of AI adoption. These align with **H2**, confirming that AI adoption significantly enhances production output, and **H4**, affirming its strong positive impact on customer satisfaction. This also supports **H7**, which hypothesizes that AI adoption positively influences customer satisfaction, given its high mean and low variability. Resource utilization (3.5315) scored high, supporting **H6** that effective resource utilization positively influences AI adoption, which was also statistically significant. Although financial sustainability (3.5064) was rated highly, it displayed only marginal significance, partially aligning with **H5** that financial sustainability positively influences AI adoption. Efficiency metrics (3.3867) exhibited positive perceptions but were not statistically significant, aligning partially with **H3** that efficiency metrics positively influence AI adoption. These results underscore the importance of overcoming barriers, leveraging resource utilization, and focusing on AI adoption's direct benefits to production output and customer satisfaction for successful integration.

Path Coefficient

Table 4 shows a path coefficient analysis, which reveals several key insights. AI adoption has a statistically significant and strong positive impact on both customer satisfaction (AA → CS; $p = 0.000$) and production output (AA → PO; $p = 0.000$), highlighting its meaningful contributions to organizational outcomes.

Table 4: Path Coefficient

No	Hypothesized paths	Original sample (O)	Sample means (M)	Standard Deviation (STDEV)	T-statistics (O/STDEV)	P
1	AA -> CS	2,9634	0,365	0,055	6,468	0,000
2	AA -> PO	3,3867	0,508	0,051	9,875	0,000
3	B -> AA	3,5064	0,295	0,079	3,661	0,000
4	EM -> AA	3,5315	0,052	0,082	0,652	0,514

5	FS → AA	2,9439	0,174	0,098	1,728	0,084
6	RU → AA	3,5262	0,215	0,105	2,063	0,039

Source: Authors

Barriers ($B \rightarrow AA$) also significantly influence AI adoption ($p = 0.000$), though the relationship is moderate, indicating that addressing barriers can improve adoption levels. Resource utilization ($RU \rightarrow AA$) shows a moderate yet significant positive effect ($p = 0.039$), emphasizing its importance in AI implementation. Efficiency metrics ($EM \rightarrow AA$), however, are not significant ($p = 0.514$), and financial sustainability ($FS \rightarrow AA$) exhibits marginal significance ($p = 0.084$), suggesting their limited direct influence on adoption. Unexpectedly, some relationships, such as barriers and financial sustainability, had negative or weak impacts, although not statistically significant. Overall, the findings underscore that addressing barriers, leveraging resources, and focusing on adoption's direct benefits (customer satisfaction and production output) can significantly enhance the effectiveness of AI integration.

Limitations

The article highlights key limitations regarding the adoption of AI in SOEs. Firstly, the article focuses on a limited and potentially homogeneous sample, which may restrict the generalizability of findings to all SOEs, particularly across different industries, sizes, or geographical regions. Additionally, survey responses may be influenced by social desirability bias or a lack of knowledge about AI, leading to potential underreporting or overreporting of barriers. Another limitation is the rapidly evolving nature of AI technology, which may render some findings outdated as new advancements emerge. The interconnectedness of barriers, such as cultural resistance and inadequate training, is another challenge, as the article may oversimplify the complexity of these issues. Moreover, the absence of regulatory framework analysis across different regions and countries could result in a lack of understanding of critical contextual factors. The study also omits enabling factors or best practices, potentially presenting an incomplete picture of AI adoption. Finally, the perspectives of AI practitioners and specialists, as well as technical considerations, are missing, limiting the depth and practicality of the findings.

Discussion

To address these limitations, future research should consider incorporating a more diverse and representative sample to improve the generalizability of findings across various SOEs. Researchers should also aim to minimize biases in survey responses by including AI practitioners or conducting in-depth interviews to gather more nuanced perspectives. Studies should integrate analyses of regional and global regulatory frameworks to better understand the contextual factors influencing AI adoption. Additionally, a focus on enabling factors and best practices that support AI adoption, alongside barriers, can provide a more balanced and holistic view. Future research should also investigate the dynamic and interconnected nature of barriers, emphasizing the relationship between cultural resistance, training gaps, and leadership support. Lastly, keeping pace with rapid technological advancements by continuously updating

findings and incorporating emerging trends will enhance the relevance and applicability of research on AI adoption in SOEs.

Conclusion

SOEs often face challenges in adopting AI due to restrictive bureaucratic systems that hinder decision-making and cultural resistance to change. Managers and employees may be reluctant to embrace AI due to a lack of understanding of its benefits or fears of job displacement. Financial constraints also prevent many SOEs from investing in necessary infrastructure and training. Additionally, the lack of qualified experts to manage AI technology, poor data management practices, and regulatory challenges further complicate AI implementation. To overcome these barriers, SOEs need to improve data management, invest in infrastructure and training, and foster a culture that values creativity and teamwork. By addressing these issues and committing to a clear AI strategy, SOEs can enhance decision-making, streamline operations, and boost productivity and competitiveness. Key actions include creating strong data governance, modernizing technology, implementing training programs, and developing compliance frameworks. This approach will position SOEs for increased efficiency and innovation.

Acknowledgements

Authors' contributions

P.M. drafted the article, which is based on the data she collected on her studies (Master of Commerce in IT Management Dissertation). P.M. was supervised by S.D. and T.M.

Ethical considerations

The ethical clearance needed to conduct this research was obtained from the College of Business and Economics (CBE) research ethics committee by the Department of Applied Information Systems at the University of Johannesburg (UJ).

Funding information

The authors received no financial support for the research, authorship, and/or publication of this article.

Data availability

The data that support the findings of this article are available from the corresponding author, P.M. upon request.

Disclaimer

The views and opinions expressed in this article are those of the authors and do not necessarily reflect the official policy or position of any affiliated agency of the authors.

Conflicts of interest

The authors declare that they have no financial or personal relationships that may have inappropriately influenced them in authoring this article

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